WISER Global Quantum+AI Program 2026 · WISER <> Nestlé

Quantum Optimization for Distributed Order Management

Choosing where to send orders a warehouse cannot fill

Data	1,109 orders · 1,110 SKUs · 8 DCs · 472 need a decision
Classical	exact MILP: **+$1.93M**, fill **90.5%** → **93.3%**, proven best in 64s
Quantum	slack-free QUBO: **18 qubits** per batch, **0** violations, validated to optimum

The problem, and why it is hard

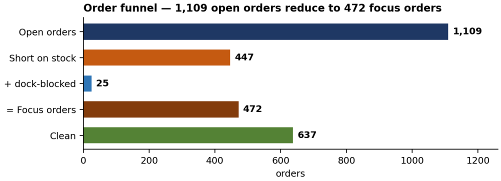
A warehouse cannot always fill an order. The planner can leave it and send a part delivery (losing sales, often paying a penalty), or move it to another DC and pay more freight.

The orders compete

- **Stock is shared** — two orders cannot both take the same SKU.
- **Capacity is shared** — docks and picking are limited each day.
- **Moves connect** — moving one order out frees stock for another to move in.

472 orders, 2,295 candidate assignments. You cannot try them all — **NP-hard in general**.

Objective = Sales − Penalty − Shipping



447 orders are short of stock. 25 more ship on a fully booked dock day. Together: **472 focus orders**.

Same rules, two searches, different answers: **37** orders moved by the solver only, 12 by the heuristic only, 26 by both.

Two ways to write the same model

Binary model (for the MILP)

$x_{ij} \in \{0,1\}$ order i from DC j; $q_{ijk} \geq 0$ cases filled;
$\Pi_i \geq 0$ penalty.

$$\max\ Z = \sum_{i,j,k} \rho_{ik} q_{ijk} - \sum_i \Pi_i - \sum_{i,j} t_{ij} x_{ij}$$

(C1) $\sum_j x_{ij} = 1$

(C2) $q_{ijk} \leq d_{ik} x_{ij}$

(C3) $\sum q_{ijk} \leq s_{jk}(u) + g_{jk}(u)$

(C4) $\sum_k q_{ijk} \geq (f_i^0 + \Gamma_i) x_{ij}$

(C5) $\sum \frac{q_{ijk}}{c_k} - \sum |S_i| x_{ij} \leq P_j$

(C6) $\sum_i x_{ij} \leq B_j$

(C7) $\Pi_i + \pi_i \sum \rho_{ik} q_{ijk} \geq M_i(1 - z_i)$

QUBO / Ising (for the quantum solver)

Rules become penalty terms in the objective:

$$H(x) = -Z(x) + \lambda_1 \sum_i \Big(\sum_j x_{ij} - 1\Big)^2 + \lambda_2 \sum_{j,k}$$

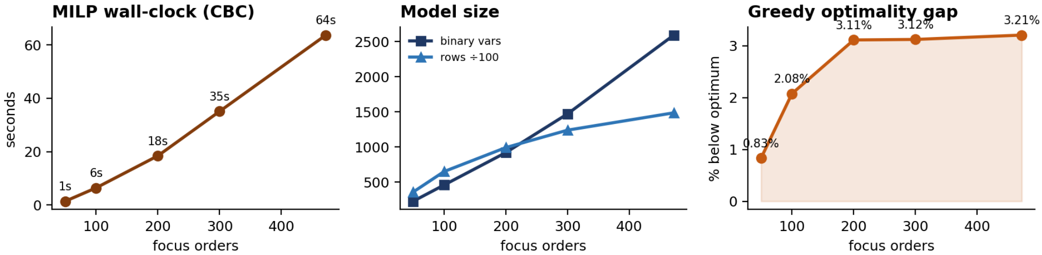
with $x_a = (1 - \sigma_a^z)/2$ and $\lambda_1 = 2 \times 10^6$.

Fill quantities are variables, not fixed.

A fixed-quantity column model scored *below* the heuristic, so it cannot serve as a ceiling. That is why we use the binary model.

Classical results, and how they scale

Method	Objective ($)	Fill	Moves	Penalty ($)	Runtime
Do nothing	44,365,994	90.47%	0	84,749	0.21s
Greedy heuristic	44,810,604	91.28%	41	75,259	0.40s
Exact MILP (PuLP + CBC)	**46,295,828**	**93.28%**	**66**	**32,333**	**63.72s**



The surprise: the solver moves 66 orders against 41, yet pays **$15,285 less** extra freight. Planning together lets it pick cheaper lanes and chain moves.

The trend: the heuristic's gap grows **0.84%** → **3.21%** from 50 to 472 orders. More orders competing means more lost by never revisiting.

Quantum: the encoding, and proving it is right

No slack qubits (Method 2)

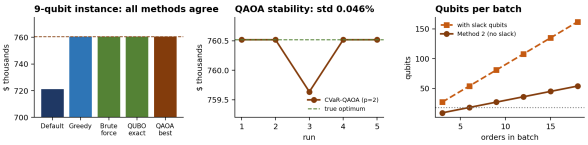
Capacity becomes a hinge penalty instead of an inequality with slack variables. Qubits then equal orders $\times$ candidate DCs exactly.

- 6 orders $\times$ 3 DCs $=$ **18 qubits**
- 18 linear, 63 quadratic terms
- a slack encoding needs $\approx$3$\times$ more

Validated against ground truth

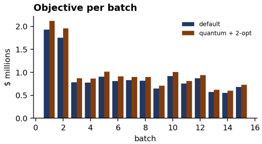
A 3$\times$3 instance has 27 feasible states, so we enumerated all of them.

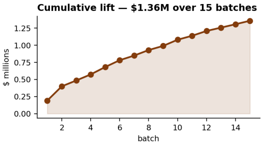
Brute force (true optimum)	760,518
QUBO ground state	760,518
QAOA best of 5	760,518
Default	721,191



QUBO energy ranking reproduces the business ranking on all 27 states — so the encoding is *faithful*, not just lucky. Five CVaR-QAOA runs ($p = 2$): mean $760,341, **std 0.046%**, 4 of 5 exactly optimal, all 5 beat the default.

Quantum at scale: batching to stay inside the hardware





90 top orders in **15 batches** of 3–6 orders $\times$ 3 DCs.

Batch size stays 9–18 qubits however many orders there are in total.

Same 75 orders	Objective	Fill
Default	$13,612,065	88.60%
Subset greedy	$14,970,736	97.27%
Quantum + 2-opt	**$14,970,736**	**97.27%**

- **+$1,358,671** (+10.0%), **0** violations, 1.07s total
- Quantum $\geq$ greedy on **15 of 15** batches
- Hybrid QAOA on a restricted master: 5 batches, gap **0.00** on every one

We report this honestly. Quantum matches the heuristic — $0 difference. At 18 qubits both find the same answer. This proves the encoding is *correct*, not that quantum is *faster*.

Limits and next steps

What limits the result

- **No demand forecast.** Nothing holds stock back at the receiving DC, so every method moves more orders than a planner would accept.
- **No throughput limit** in the data — we use peak observed use instead.
- **Quantum** covers 75 orders, on noise-free simulation. Real hardware adds gate errors, and a QUBO only *discourages* a broken rule, never forbids it.
- **Batching** drops the links between batches — the very coupling that gave the classical solver its freight saving.

Next steps

1. **Add the demand forecast.** Without it all stock looks free to use, so the model spends stock the receiving DC needs later.
2. **Review which DCs carry which SKUs.** A missing SKU blocks 40% of candidate moves.
3. **Revisit the 5% gate.** It blocks 46% of moves. It is a policy setting, not a law of the warehouse.

Most moves are blocked by *policy*, not by buildings: docks and picking together block only 0.8% of candidates.